MANDAR JOSHI

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PROFESSIONAL SUMMARY

Software engineer and B.Tech IT student contributing to **Supabase, FOSSology (Linux Foundation), and Elixir**, with fixes shipped in auth security, API performance, memory safety, and developer tooling. Builds distributed systems and production platforms in Go, Elixir, C, PHP, and TypeScript.

EDUCATION

Vellore Institute of Technology — Vellore, India
B.Tech in Information Technology — *Jul. 2024 – May 2028 (Expected)*
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Cloud Computing

TECHNICAL SKILLS

Languages: Go, Elixir, TypeScript, C, PHP, SQL, JavaScript, Bash
Frameworks: Phoenix, Node.js, Express.js, Next.js, React.js, React Native, gRPC/Protobuf, Tailwind CSS
Infra/Tools: Docker, Docker Compose, Kubernetes, Linux, Nginx, Caddy, Cloudflare, GitHub Actions, CI/CD
Databases/Other: PostgreSQL, Supabase, Redis, Apache Kafka, ExUnit, PHPUnit, Git, Postman

OPEN SOURCE

SUPABASE INC. — *15 Merged PRs, 25 Issues*
- Secured self-hosted Realtime by moving hardcoded encryption keys to configurable `.env` variables, and added strict JWT audience/issuer validation to the Supabase Server.
- Fixed self-hosted OAuth discovery by proxying `.well-known` routes through the Caddy/Nginx API gateway, resolving 404s that broke auth.
- Stabilized Docker Compose startup by adding healthcheck grace intervals that stopped cascading service failures.
- Fixed test flakes and log-suppression bugs in the Elixir-based Realtime backend.

FOSSOLOGY (LINUX FOUNDATION) — *11 Merged PRs, 8 Issues*
- Cut database round-trips in the `/scanned` API endpoint by eliminating N+1 queries and redundant lookups, and fixed null-pointer crashes; brought member API endpoints in line with REST conventions.
- Fixed SQL formatting bugs and type misuse in the DAO layer, covering each fix with PHPUnit regression tests.
- Fixed memory-safety bugs in the C-based `pkgagent`: replaced unsafe `strcpy()` calls, fixed NULL-before-free crashes, and corrected string-length tracking.

ELIXIR (ELIXIR-LANG) — *6 Merged PRs, 4 Issues*
- Made `mix format` output stable for comments after operators and inside tuple arguments, with regression tests.
- Fixed the formatter dropping parentheses on anonymous calls with `do-end` blocks, which emitted unparseable code.

STAMPEDE (CREATED LIBRARY) | LINK — *Creator & Maintainer, Published on Hex*
- Created an open-source Elixir load-testing library for Phoenix that speaks the Channels and LiveView protocols natively, plus HTTP and WebSockets, running every virtual user as a BEAM process.
- Sustained 600K concurrent users on AWS by spreading load across nodes via Erlang distribution; used it to load test Supabase Realtime, with CI thresholds, a live dashboard and run-to-run regression checks.

WORK EXPERIENCE

SOFTWARE DEVELOPER INTERN | ROBOTICS DATA STARTUP — *Aug. 2026 – Present*
- Building web/mobile apps (Next.js, React Native, Supabase, Cloudflare) used by 40+ field operators to manage egocentric video datasets for robotics and embodied AI.
- Built a video ingestion and processing pipeline handling 150+ hours of footage per week, cutting dataset turnaround from 2 days to under 6 hours.
- Built QA tooling that flags blurry frames and missing metadata before delivery, cutting manual review time by 35%.

SOFTWARE ENGINEER INTERN | B2B SAAS STARTUP — *Dec. 2025 – Jun. 2026*
- Built 20+ REST APIs in Node.js/Express for an enterprise operations platform serving 50K+ requests/day.
- Shipped a workflow automation engine with AI-assisted actions, adopted by 3 client teams.
- Moved report generation to a queue-driven design, cutting p95 API latency from 4.2s to 650ms.

PROJECTS

Resync: UPI Autopay Recovery Platform | LINK — *2026*
- Architected a **9-microservice** payment recovery platform in **Go + gRPC**, with a dunning engine that classifies UPI failure codes and schedules retries based on the failure reason.
- Built an event-driven pipeline on **Kafka** (partition-keyed ordering), **Redis** distributed locks, and idempotency keys to prevent **double-charges**, verified with 500 concurrent duplicate retry events in load tests.

Continum: AI/ML Workflow Automation Platform | LINK — *2026*
- Built no-code AI workflows that draft context-aware Gmail replies and schedule meetings, processing 1,000+ emails across pilot users.
- Deployed as microservices on Kubernetes with GitHub Actions CI/CD running 120+ tests on every push.